# MWEZICARE AI

Understand your cycle. Recognize the signs. Get care sooner.

## AI-Powered Menstrual Health & Endometriosis Early-Recognition and Referral Platform

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| **Proposed focus**<br>Girls and young women in Kenya |
| **Core approach**<br>Symptom tracking • Clinically reviewed triage • Location-aware referrals |
| **Proposed technology**<br>Conversational AI • Pattern monitoring • Verified facility directory |

Prepared as a concept brief for discussion with a technology partner and potential development for a femtech innovation funding application.

# 1. Executive Summary

MweziCare AI is a proposed mobile-first menstrual health companion for adolescent girls and young women in Kenya. It aims to help users understand menstrual health, identify symptoms that warrant medical assessment, track changes over time, and connect with appropriate healthcare services sooner.

The platform would combine structured symptom tracking, clinically reviewed decision-support rules, a conversational health assistant, and location-aware referral navigation. It would focus on early recognition of warning signs associated with endometriosis and other menstrual or pelvic health conditions—not on making an automated diagnosis.

**Core positioning: MweziCare AI turns menstrual health information into an actionable pathway from awareness, to monitoring, to appropriate clinical care.**

# 2. The Problem

Many girls and young women are taught to normalize severe period pain. Some miss school, work, sleep, or everyday activities without realizing that persistent or debilitating pain deserves medical evaluation. Endometriosis may cause significant menstrual pain, pelvic pain outside menstruation, pain with bowel movements or urination, pain during sex, heavy bleeding, fatigue, and—in some people—difficulty becoming pregnant. These symptoms can overlap with other conditions, so symptoms alone cannot confirm endometriosis.

The gap MweziCare AI seeks to address is delayed recognition, limited access to understandable menstrual health information, and difficulty finding suitable gynaecological care.

# 3. Proposed Solution

MweziCare AI would offer four connected functions:

• **Menstrual symptom tracking:** Period dates, bleeding, pain severity (0–10), pain location, duration, fatigue, nausea, bowel or urinary symptoms, and impact on daily activities.

• **AI-supported pattern monitoring:** Identify recurring, worsening, or disruptive symptoms and flag patterns that warrant clinical assessment.

• **Conversational health education:** Answer common menstrual health questions in accessible language, with potential English and Kiswahili support, using reliable and clinician-reviewed content.

• **Location-aware care navigation:** With permission, help users find nearby verified facilities that offer gynaecology services, with contact information, directions and referral options.

**The app should explain why it is suggesting a next step, communicate uncertainty, and never claim that a user has—or does not have—endometriosis based only on app data.**

# 4. How the App Would Work

**Step 1 — Create a profile and log symptoms.** The user records cycle dates, pain severity, bleeding patterns, associated symptoms and how symptoms affect school, work, sleep and daily activities.

**Step 2 — Review patterns.** Clinically reviewed rules and, when appropriately validated, AI models examine changes over time, including recurring severe pain, pain between periods and symptoms associated with bowel movements or urination.

**Step 3 — Receive a safe next-step recommendation.** Depending on the reported symptoms, the app may recommend routine tracking, booking a clinical assessment soon, or seeking urgent medical care. The app must not delay urgent care while gathering more data.

**Step 4 — Find appropriate care.** If the user consents to location access, the app shows nearby facilities and distinguishes confirmed gynaecology services from general outpatient services.

**Step 5 — Share a symptom summary.** The user may choose to share a concise history of symptoms with a qualified clinician to support the consultation.

# 5. Menstrual Health Education: What Is Common and What Needs Attention?

| Experience | Suggested app guidance |
|---|---|
| Mild or moderate cramps that improve and do not prevent normal activities | Often common. Offer self-care information and routine tracking. |
| Occasional changes in cycle timing | Record changes; recommend assessment if they persist or are concerning. |
| Pain that repeatedly prevents school, work, sleep or daily activities | Recommend clinical assessment. Do not dismiss as normal period pain. |
| Worsening cramps, pelvic pain between periods, or pain with bowel movements or urination | Flag for clinical evaluation, including consideration of endometriosis and other causes. |
| Unusually heavy or prolonged bleeding, or repeated bleeding between periods | Recommend assessment to identify the cause. |
| Sudden severe pelvic pain, fainting, or severe pain with possible pregnancy | Seek urgent medical care; do not wait for an app assessment. |

These are educational triage examples, not diagnostic criteria. Symptoms can have multiple causes. Clinical advice should be reviewed by qualified health professionals and aligned with local clinical guidance.

# 6. AI and Technical Design

**Symptom intelligence:** Start with transparent, clinician-reviewed rules for triage and escalation. Consider machine learning only after there is sufficient, appropriately consented, representative data and independent validation.

**Conversational assistant:** Use a curated medical knowledge base and constrained responses to answer common questions. Include safeguards for urgent symptoms, clear uncertainty, and referral to human care where needed.

**Longitudinal monitoring:** Detect symptom recurrence and changes over time. Present patterns and reasons for concern rather than unsupported diagnostic labels or numerical probabilities that have not been validated.

**Privacy and security:** Minimize data collection, obtain informed consent, use appropriate access controls and encryption, define retention and deletion policies, and apply age-appropriate protections. Sensitive reproductive health data require particular care.

# 7. Kenya-Specific Location and Referral Feature

The app could use the Kenya Master Health Facility List as a starting source for facility discovery, then verify which facilities actually provide gynaecology services. A directory entry alone is not proof that a specialist is available or accepting appointments.

• Ask permission before accessing the user's location; offer manual county, town or area search as an alternative.

• Display facility name, location, phone number, confirmed services, opening details where verified, and directions.

• Distinguish general outpatient services from facilities with verified gynaecology care.

• Show costs, insurance acceptance, appointment availability and referral requirements only when reliable information is available.

• Provide safe fallback guidance when no nearby specialist can be verified, including seeking assessment at an appropriate local facility.

**Illustrative user journey: a young woman in Thika logs severe period pain over several cycles. The app flags repeated disruption to daily life and explains that clinical assessment is warranted. With consent, it lists nearby facilities with verified gynaecology services and lets her contact a facility or share a symptom summary.**

# 8. Alignment with UNICEF Femtech Ventures

**Frontier technology:** AI-supported symptom monitoring and referral decision support.

**Girls' and women's health:** Age-appropriate menstrual health education and earlier recognition of symptoms that warrant care.

**Local relevance:** Kenya-specific facility navigation, accessible language, and consideration of low-connectivity settings.

**Open-source approach:** Publish appropriate core components under a suitable open-source licence, with clear documentation and responsible handling of any sensitive data.

**Measurable impact:** Evaluate changes in knowledge, appropriate care-seeking, referral completion where verifiable, usability and safety.

Funding programme fit must be confirmed against the official call requirements. The screenshot describes up to US$100,000 in funding and 12 months of mentorship. The previously noted deadline was 4 November 2026; verify current details directly with UNICEF before submission.

# 9. Proposed Minimum Viable Product (MVP)

| Module | First-version scope |
|---|---|
| 1. Menstrual profile | Cycle dates, symptom logs, pain score, bleeding and impact on daily activities. |
| 2. Symptom assessment | Clinician-reviewed rules, clear explanations, escalation for urgent symptoms, endometriosis warning-sign education. |
| 3. Care navigation | Facility search, permission-based location, verified gynaecology-service filters, directions and contact details. |
| 4. Clinical summary | A user-controlled symptom report that can be shared with a clinician. |
| 5. Impact dashboard | Aggregated indicators for usage, symptom flags and referral follow-up where verified, without exposing identifiable data. |

# 10. Pilot and Impact Measurement

Begin with a small, ethically designed pilot with consenting users and qualified clinical oversight. Test usability, language, safety, accessibility and whether recommendations are understandable. Do not use the tool as a substitute for professional diagnosis or treatment.

• Menstrual health knowledge before and after using the app.

• Proportion of users receiving a recommendation for assessment who subsequently attend care, where follow-up can be verified with consent.

• Time from recognition of a concerning symptom pattern to seeking care.

• Appropriateness and safety of triage recommendations, including urgent escalation.

• User trust, accessibility, privacy incidents and satisfaction.

Avoid claiming that the product reduces diagnostic delays or improves clinical outcomes until the pilot provides evidence.

# 11. Suggested Team Responsibilities

**Clinical/product lead:** Define the user problem, clinical content, symptom categories, referral workflow, safeguarding and pilot outcomes. A nursing background can be valuable here, with a qualified gynaecologist and other appropriate clinicians providing oversight.

**Technology partner:** Lead architecture, mobile/web development, AI integration, data security, facility search, location features and technical testing.

**Joint responsibilities:** User research, consent and privacy design, open-source strategy, partnership development, funding application and impact reporting.

# 12. Key Risks and Safeguards

| Risk | Safeguard |
|---|---|
| False reassurance or missed urgent condition | Conservative escalation rules, urgent-care messaging, clinical validation and safety testing. |
| False alarms or unnecessary referrals | Explain uncertainty, refine rules with clinicians, monitor referral appropriateness. |
| Outdated facility information | Verify services and contacts regularly; show when information was last confirmed. |
| Sensitive data exposure | Data minimization, encryption, access controls, consent, retention limits and age-appropriate safeguards. |
| Biased or unvalidated AI | Start with transparent rules; evaluate across age groups, languages and settings before expanding. |

# 13. Draft Message to the Technology Partner

I've been thinking about an AI-powered femtech solution that we could develop together and potentially submit to the UNICEF Femtech Ventures funding programme.

**Proposed idea: MweziCare AI — an AI-powered menstrual health and endometriosis early-detection and referral platform for girls and young women in Kenya.**

The problem I want us to address is how many girls and women are taught to normalize severe period pain without knowing when their symptoms require medical attention. Conditions such as endometriosis can take years to recognize, and access to appropriate gynaecological care can be challenging.

The idea is to build a mobile-first application that allows users to track their menstrual cycles, pain severity, bleeding patterns and associated symptoms over time. Using AI and clinically reviewed assessment rules, the app would identify concerning symptom patterns, educate users about possible warning signs of endometriosis, and recommend an appropriate next step.

A key feature would be location-based healthcare navigation. With the user's consent, the app would identify nearby verified health facilities in Kenya that offer gynaecology services, provide directions and contact information, and eventually support referrals and appointment booking through partner facilities.

I would also like us to explore a conversational AI assistant that answers menstrual health questions in accessible language, including Kiswahili, while protecting users' privacy.

The app would not claim to diagnose endometriosis. Its role would be early symptom recognition, risk-informed triage, monitoring and connection to qualified healthcare professionals.

For the initial prototype, I envision four core features: menstrual symptom tracking, AI-supported symptom assessment, a Kenyan health facility locator, and a dashboard for measuring impact.

I believe the concept could align with UNICEF's focus on AI-driven, locally developed and open-source innovations for girls' and women's health. We would need to establish a functional prototype, clinical validation, appropriate data protection and a measurable pilot.

I'd like us to assess the technical feasibility, architecture, prototype timeline, open-source approach and how we could turn this into a credible funding application before the 4 November 2026 deadline.

What do you think, and how would you approach building the MVP?

# 14. Recommended Next Steps

• Confirm the funding call's current eligibility criteria, deadline, company registration requirements, prototype expectations and open-source conditions.

• Agree on the target age group and conduct user discovery with girls and young women, with appropriate consent and safeguarding.

• Recruit a qualified gynaecologist or clinical advisory group to review symptom guidance and triage pathways.

• Map and verify pilot referral facilities in one initial Kenyan county or town.

• Build and test a functional MVP, including privacy, accessibility and urgent-symptom safeguards.

• Define pilot measures and a realistic implementation budget before submitting the application.

### Closing recommendation

**Position MweziCare AI as an early-recognition and care-navigation tool within broader menstrual health—not as an AI that diagnoses endometriosis. This is a more clinically responsible, testable and credible starting point. Confirm whether the team has a registered company and working prototype, as these may be important to funding eligibility.**

## Reference and verification note

This document is a concept proposal for discussion, not an official UNICEF document and not evidence of programme acceptance. Before external submission, verify programme details on the official UNICEF Femtech Ventures call page and obtain clinical and legal review of the proposed product.

Official call page: https://www.unicef.org/innovation/FemTechCall